

LUGANO, Switzerland

WMO: 067700

Lat: 46.0039N

Lon: 8.9601E

Elev: 915

StdP: 14.22

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 27.8	(c) 29.9	(d) 1.6	(e) 6.2	(f) 40.5	(g) 7.0	(h) 8.1	(i) 43.9	(j) 21.0	(k) 47.1	(l) 18.7	(m) 45.9	(n) 3.2	(o) 340	(p) 0.390

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 16.8	(c) 86.6	(d) 71.0	(e) 84.2	(f) 69.8	(g) 82.1	(h) 68.8	(i) 73.3	(j) 84.1	(k) 71.8	(l) 81.7	(m) 70.4	(n) 79.9	(o) 5.4	(p) 190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 69.7	(b) 113.0	(c) 80.0	(d) 68.2	(e) 107.0	(f) 78.1	(g) 66.8	(h) 102.1	(i) 76.6	(j) 37.4	(k) 84.4	(l) 36.0	(m) 81.9	(n) 34.8	(o) 80.1	(p) 79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 17.8	(b) 14.5	(c) 11.7	(e) 23.1	(f) 90.5	(g) 3.8	(h) 2.4	(i) 20.4	(j) 92.2	(k) 18.1	(l) 93.6	(m) 16.0	(n) 95.0	(o) 13.2	(p) 96.7		
			DB													
			WB													
				20.8	75.8	3.4	1.9	18.4	77.2	16.4	78.4	14.5	79.5	12.1	80.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.7	39.0	41.3	48.5	54.4	61.9	69.3	72.9	71.8	64.4	56.6	47.2	40.4
	DBStd	12.91	4.29	4.80	5.65	5.52	5.27	5.47	4.48	4.37	4.95	5.02	5.15	4.63
	HDD50	1118	342	244	95	20	1	0	0	0	0	8	109	299
	HDD65	4075	806	662	513	320	123	20	3	4	67	261	533	763
	CDD50	3210	1	2	47	151	370	580	709	676	433	214	26	1
	CDD65	695	0	0	0	1	28	150	247	216	51	2	0	0
	CDH74	4351	0	0	5	11	139	953	1739	1313	183	8	0	0
	CDH80	973	0	0	0	3	13	212	430	301	13	1	0	0
Wind	WSAvg	3.6	3.2	3.5	4.4	4.3	3.9	3.9	3.9	3.6	3.5	3.0	3.2	3.1
Precipitation	PrecAvg	62.6	3.0	2.8	3.7	5.5	7.6	6.5	6.0	6.4	6.7	5.9	5.7	2.8
	PrecMax	109.7	11.0	10.9	12.4	20.2	16.8	17.5	15.6	19.3	17.6	21.3	23.1	7.8
	PrecMin	38.5	0.0	0.0	0.1	0.4	1.9	0.7	0.6	0.5	0.4	0.0	0.0	0.0
	PrecStd	14.5	2.6	2.5	2.8	3.5	3.5	3.5	3.7	3.7	4.8	4.6	4.7	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	59.8	62.4	72.1	74.7	81.5	88.8	89.3	90.0	81.6	73.8	62.3	57.5
		MCWB	45.2	45.9	51.3	57.5	64.7	70.8	72.0	72.8	68.5	61.8	51.1	42.2
	2%	DB	51.8	55.5	65.6	70.5	77.8	85.2	86.5	86.1	78.8	69.5	58.9	52.2
		MCWB	40.1	43.0	50.1	55.5	62.8	70.0	71.1	71.4	66.6	60.5	51.3	43.1
	5%	DB	48.2	51.8	61.6	67.5	74.7	82.2	84.4	83.1	75.8	66.5	56.4	49.2
		MCWB	39.8	42.1	48.4	54.3	61.2	68.7	70.2	69.6	64.7	58.9	50.4	43.4
	10%	DB	45.5	48.8	58.2	64.4	71.6	79.5	82.3	80.8	73.1	64.3	54.6	47.3
		MCWB	40.0	40.8	47.1	52.7	59.8	67.1	68.9	68.5	63.3	58.3	50.0	42.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.0	48.4	54.0	60.3	67.5	74.6	75.5	75.0	70.6	64.5	56.2	49.3
		MCDB	54.7	57.2	66.5	71.3	77.7	85.0	85.4	86.6	79.4	70.0	58.3	51.2
	2%	WB	44.2	45.8	51.9	57.9	65.1	72.0	73.5	72.9	68.5	62.7	54.2	46.7
		MCDB	47.9	51.9	61.3	67.3	74.9	82.0	84.2	83.7	76.5	67.1	56.4	48.8
	5%	WB	42.6	44.0	50.3	56.1	63.1	70.3	71.9	71.4	66.9	61.0	52.9	45.0
		MCDB	45.6	48.5	58.2	64.2	71.9	80.1	81.9	81.2	73.7	64.9	55.1	47.5
	10%	WB	41.1	42.6	48.9	54.5	61.4	68.6	70.5	70.0	65.1	59.4	51.2	43.5
		MCDB	44.2	46.6	55.8	61.6	69.1	77.7	80.2	78.9	71.2	63.1	53.5	46.5
Mean Daily Temperature Range	5% DB	MDBR	11.5	13.4	15.7	15.0	15.2	16.2	16.8	16.2	14.9	12.6	11.1	10.8
		MCDBR	16.9	19.4	23.1	20.8	20.2	19.1	19.0	19.5	18.1	16.2	14.0	13.7
		MCWBR	9.7	10.9	11.3	9.8	9.9	9.5	9.6	9.5	9.5	9.6	8.9	8.4
	5% WB	MCDBR	13.0	16.3	19.7	16.8	17.3	17.3	17.4	17.4	15.9	13.0	9.9	10.2
		MCWBR	8.5	10.0	10.5	8.3	9.1	9.2	9.6	9.1	8.8	8.0	7.1	7.1
Clear-Sky Solar Irradiance	taub		0.322	0.359	0.417	0.477	0.484	0.502	0.484	0.475	0.458	0.428	0.364	0.319
	taud		2.308	2.233	2.130	2.014	2.047	2.052	2.115	2.149	2.153	2.188	2.318	2.361
	Ebn at Noon		241	253	253	248	251	246	249	246	238	224	223	230
	Edh at Noon		25	33	42	52	52	52	48	45	41	34	25	22
All-Sky Solar Radiation	RadAvg		486	766	1170	1423	1638	1907	1926	1648	1258	786	476	404
	RadStd		65	99	159	196	148	139	141	122	104	108	78	55

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.86	+1.57	N/A	N/A	N/A	N/A	-148	-283	+277	+151
Regional (109 neighbors)	N/A	N/A	N/A	+1.30	+0.74	+0.58	N/A	N/A	+106	+29

Nomenclature: See separate page